CASE STUDY 18

LIBRARY BOOK BORROWING AND RETURN SYSTEM

Python Program

```
print("===== LIBRARY MANAGEMENT SYSTEM =====")

books = [
    ("Python Programming", 3),
    ("Data Structures", 2),
    ("Artificial Intelligence", 4)
]

borrowed = {}

while True:
    print("\n===== MENU =====")
    print("1. View Available Books")
    print("2. Borrow a Book")
    print("3. Return a Book")
    print("4. Exit")

    choice = input("Enter your choice: ")

    if choice == "1":
        print("\n===== AVAILABLE BOOKS =====")
```

```
    for title, copies in books:
        print("{}: {} copies available".format(title, copies))

elif choice == "2":
    title = input("Enter the book name to borrow: ")

    found = False

    for i in range(len(books)):
        if books[i][0].lower() == title.lower():
            found = True
            book_title, copies = books[i]

            if copies > 0:
                books[i] = (book_title, copies - 1)
                borrowed[book_title] = borrowed.get(book_title, 0) + 1
                print("Book borrowed successfully.")
            else:
                print("Book is currently unavailable.")

            break

    if not found:
        print("Book not found in the library.")
```

```
elif choice == "3":
    title = input("Enter the book name to return: ")

    borrowed_title = None

    for book_title in borrowed:
        if book_title.lower() == title.lower():
            borrowed_title = book_title
            break

    if borrowed_title is not None and borrowed[borrowed_title] > 0:
        for i in range(len(books)):
            if books[i][0] == borrowed_title:
                book_title, copies = books[i]
                books[i] = (book_title, copies + 1)
                borrowed[borrowed_title] -= 1

                print("Book returned successfully.")
                break
    else:
        print("No borrowed copy of this book was found.")

elif choice == "4":
    print("\nThank you for using the library system.")
    break
```

```
    else:
        print("Invalid choice. Please try again.")

print("Program completed.")
```

Output:

